Complete Pass-Through in Levels

Kunal Sangani

February 2026

Disclaimer

This presentation contains my own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the author and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

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Beige Book 101

February 14, 2024

What is the Beige Book?

The Beige Book is a report that offers a snapshot of economic developments in each of the Fed's 12 regional Reserve Bank

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The Kansas City Fed section of the Beige Book covers nine topics: labor markets; prices; consumer spending; community conditions; manufacturing and other business activity; real estate and construction; community and regional banking;

What information do we collect?

- **Prices:** Characterizes changes in the prices consumers pay as well as price changes businesses face for their inputs and raw materials (e.g., steel, oil, corn, grain and lumber). Assesses the pass-through of production costs to customers and expectations about prices pressures.

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WSJ The Wall Street Journal

How Much Do Tariffs Raise Prices? - WSJ

How much tariffs raise prices is complicated because various factors affect the "pass-through" of such duties to customers.

Mar 2, 2025

What information do we collect?

- **Prices:** Characterizes changes in the prices consumers pay as well as price changes businesses face for their inputs and raw materials (e.g., steel, oil, corn, grain and lumber). Assesses the pass-through of production costs to customers and expectations about prices pressures.

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Basic Approach

Consider an environment in which the U.S. levies a tariff of rate τ_i on country i and Δτ_i reflects the change in the tariff rate. Let ε<0 represent the elasticity of imports with respect to import prices, let φ>0 represent the passthrough from tariffs to import prices, let m_i>0 represent total imports from country i, and let x_i>0 represent total exports. Then the decrease in imports due to a change in tariffs equals Δτ_i*ε*φ*m_i<0. Assuming that offsetting exchange rate and general equilibrium effects are small enough to be ignored, the reciprocal tariff that results in a bilateral trade balance of zero satisfies:

$$\Delta\tau_i = \frac{x_i - m_i}{\varepsilon * \varphi * m_i}.$$

Pass-Through in Logs and Levels

- Incomplete long-run pass-through of input cost changes.

  E.g., Goldberg and Hellerstein (2008), Nakamura and Zerom (2010), De Loecker et al. (2016), Hong and Li (2017).

  - When costs increase 10%, firms raise prices $<$10%.
  - Incomplete even at long horizons and after accounting for input cost share.
  - Prevailing explanation: Curvature of residual demand.

Pass-Through in Logs and Levels

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  - When costs increase 10%, firms raise prices $<$10%.
  - Incomplete even at long horizons and after accounting for input cost share.
  - Prevailing explanation: Curvature of residual demand.
- Today: Measure pass-through on a dollars-and-cents basis.
- Result: Firms exhibit complete pass-through in levels.
  - For persistent, industry-wide shocks: $1/unit cost increase leads to $1/unit higher prices.
  - Explains incomplete "log pass-through" and variation across products/firms.
  - Pattern spans many markets (gas stations, food products, manufacturing industries).

Explaining Pass-Through in Levels

- Workhorse macro models predict pass-through in levels equals markup, $\mu > 1$.
  - CES, or any homothetic industry demand system given industry-wide cost shock.
    (E.g., Kimball 1995; Atkeson and Burstein 2008; HSA from Matsuyama and Ushchev 2022.)
  - Reason: Nash-Bertrand conduct + scale-invariant demand.

Explaining Pass-Through in Levels

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  - Reason: Nash-Bertrand conduct + scale-invariant demand.
- How to reconcile price *dynamics* that resemble perfect competition with imperfectly competitive price *levels*?
- One solution: Shift-invariant demand systems.
  - Many familiar from IO literature (e.g., nested/mixed logit, probit, address models).
  - Preserves assumptions on firm conduct, link between markups and demand elasticity.
  - Preserves neutrality with respect to aggregate price level.

Macro Implications

1. Asymmetry, size-dependence, and heterogeneity in “log pass-through.”
2. Mechanical effects of cost shocks on relative price and markup dispersion.
3. In industry model: Dynamics of industry gross margins, profits, and entry.
4. In multisector model: Differential pass-through of materials vs. labor costs. (Okun 1981)
5. In input-output model: Low volatility of consumer inflation relative to commodity prices.
6. Surge in “cheapflation” and inflation inequality in 2021–2023.

Selected Related Literature

- **Theoretical and empirical determinants of pass-through**:
  - E.g., Bulow and Pfleiderer (1983), Campa and Goldberg (2005), Burstein et al. (2006), Nakamura and Zerom (2010), Weyl and Fabinger (2013), Burstein and Gopinath (2014), Hong and Li (2017), Mrázová and Neary (2017), Amiti et al. (2019), Conlon and Rao (2020), Butters et al. (2022), Minton and Wheaton (2022), Miravete et al. (2023).
  - In paper: Appendix Table A1 collects studies measuring pass-through in levels and logs.
  - Focus on long-run pass-through of industry-wide ("common") shocks.
    Abstract from (1) speed of adjustment, (2) firm-specific shocks.
- **Demand systems**:
  - Surveys by Anderson et al. (1992), Berry and Haile (2021), Gandhi and Nevo (2021).
  - Selected related work: Sattinger (1984), McFadden (1981), Perloff and Salop (1985), Berry et al. (1995), Atkeson and Burstein (2008), Matsuyama and Ushchev (2017), Griffith et al. (2018), Miravete et al. (2023), Birchall et al. (2024), Barro (2024).

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Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline
Commodity	$1
Other components of marginal cost	$1
Total marginal cost	$2
Price	$4

Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline	
Commodity	$1	+$0.20
Other components of marginal cost	$1	
Total marginal cost	$2	+$0.20
Price	$4	

Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline		New
Commodity	$1	+$0.20	$1.20
Other components of marginal cost	$1		$1.00
Total marginal cost	$2	+$0.20	$2.20
Price	$4	?	?

Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline		New	*% Change*
Commodity	$1	+$0.20	$1.20	*+20%*
Other components of marginal cost	$1		$1.00	
Total marginal cost	$2	+$0.20	$2.20	*+10%*
Price	$4	?	?	

Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline		New	*% Change*
Commodity	$1	+$0.20	$1.20	*+20%*
Other components of marginal cost	$1		$1.00	
Total marginal cost	$2	+$0.20	$2.20	*+10%*
Price	$4	+$0.40	$4.40	*+10%*

- Complete pass-through in logs / fixed percentage markup:
  $p = \mu(c + w) \;\Rightarrow\; \Delta p = \mu \Delta c.$

Pass-Through in Levels: Example

- Leontief production in commodity ($1/unit) and other variable costs ($1/unit).

Cost per unit	Baseline		New	*% Change*
Commodity	$1	+$0.20	$1.20	*+20%*
Other components of marginal cost	$1		$1.00	
Total marginal cost	$2	+$0.20	$2.20	*+10%*
Price	$4	+$0.20	$4.20	*+5%*

- Complete pass-through in logs / fixed percentage markup:
  $p = \mu(c + w) \;\Rightarrow\; \Delta p = \mu \Delta c.$
- Complete pass-through in levels / fixed "additive markup":
  $p = (c + w) + m \;\Rightarrow\; \Delta p = \Delta c.$
  Appears incomplete in logs, even relative to cost share.

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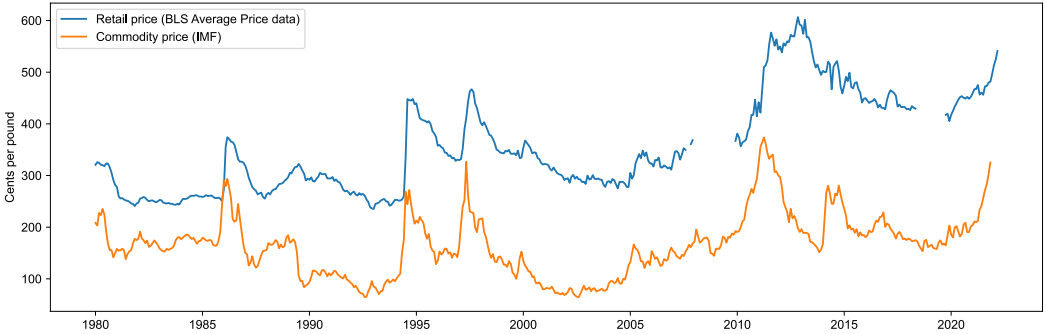
Explaining Pass-Through in Levels

Implications

Pass-through for six staple food products

- BLS tracks price levels for staple food products (Average Price Data program).
- Identify six products where we can match units and prices of commodity inputs.
  - Coffee, sugar, beef, rice, flour, frozen orange juice concentrate.

Figure: Arabica coffee commodity costs (IMF) and retail ground coffee prices (BLS).



Canonical approach to measure pass-through of cost changes

- Specification à la Campa and Goldberg (2005), Nakamura and Zerom (2010), etc.

- Price change at time t in market m due to commodity cost changes in last K periods:

$$\Delta p_{m,t} = a_m + \sum_{k=0}^{K} b_k \Delta c_{m,t-k} + \varepsilon_{m,t}.$$

  Long-run pass-through is $\sum_{k=0}^{K} b_k$.

- Suppose $p = \mu(c + w) + m$. Then $\sum_{k=0}^{K} b_k \to \mu$ if...
  - The commodity price c is unit root (autocorrelation coefficients 0.967–0.997).
  - K large enough (we use $K = 12$ months).
  - No reverse causality / anticipation (we ensure one-way Granger causality from Δc to Δp and show that leads of Δc do not predict Δp).

Example: Pass-through of coffee commodity costs to CPI

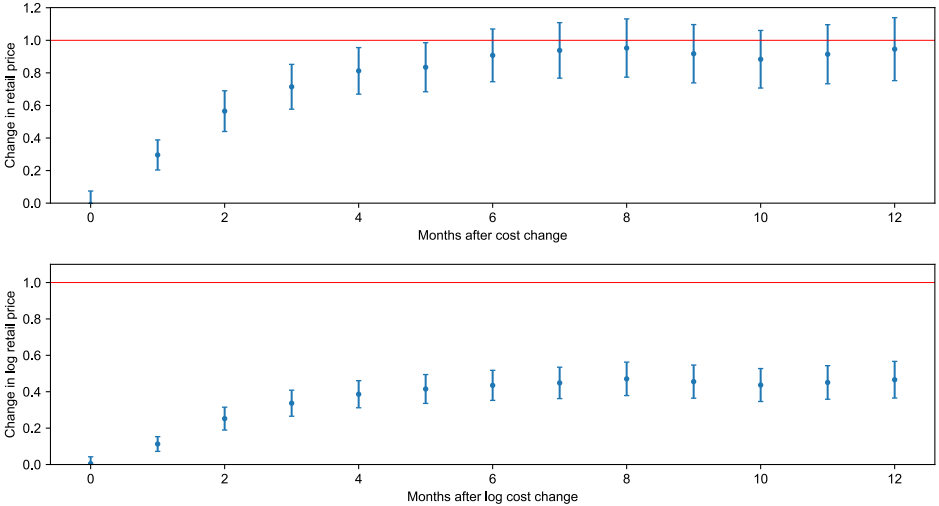


Figure: Passthrough in levels (top) and in logs (bottom).

Pass-through for six food products matched to commodities

Commodity (IMF)	Final Good (BLS)
Arabica coffee price, per lb.	Coffee, 100%, ground roast
Sugar, No. 16, per lb.	Sugar, white, per lb.
Beef, global price, per lb.	Ground beef, 100% beef
Rice, Thailand, per metric ton	Rice, white, long grain, uncooked
Wheat, global price, per metric ton	Flour, white, all purpose
Frozen orange juice solids, per lb.	Orange juice, frozen concentrate

- Monthly commodity prices (IMF), retail prices (BLS Average Price Data), 1990–2020.
- Match units (e.g., lbs flour per bushel of wheat) using USDA conversion tables.

Pass-through for six food products matched to commodities

Commodity (IMF)	Final Good (BLS)	Pass-through (12 mos.) Logs		Levels	
Arabica coffee price, per lb.	Coffee, 100%, ground roast	0.466	(0.051)	**0.946**	**(0.099)**
Sugar, No. 16, per lb.	Sugar, white, per lb.	0.370	(0.035)	0.691	(0.072)
Beef, global price, per lb.	Ground beef, 100% beef	0.410	(0.068)	**0.899**	**(0.126)**
Rice, Thailand, per metric ton	Rice, white, long grain, uncooked	0.307	(0.049)	**0.882**	**(0.169)**
Wheat, global price, per metric ton	Flour, white, all purpose	0.240	(0.048)	**0.865**	**(0.160)**
Frozen orange juice solids, per lb.	Orange juice, frozen concentrate	0.327	(0.040)	**0.974**	**(0.111)**

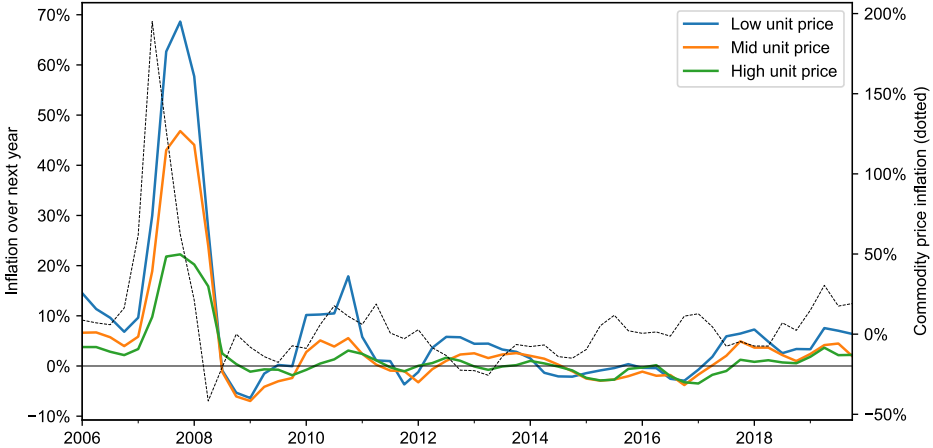
- Monthly commodity prices (IMF), retail prices (BLS Average Price Data), 1990–2020.
- Match units (e.g., lbs flour per bushel of wheat) using USDA conversion tables.
- **Cannot reject complete pass-through in levels for 5 of 6.** (Reject in logs for all.)

Exploiting cross-sectional variation

- Complete pass-through in levels across different products.
- Two exercises exploiting cross-sectional variation in NielsenIQ Retail Scanner data.
- Explains heterogeneity in log pass-through across products (UPCs):
  - High-priced products in rice/coffee/flour have lower “log pass-through.”
  - Explained by same pass-through in levels across products.
- Explains heterogeneity in log pass-through across retailers selling same UPC:
  - Retailers selling same UPC at higher price have lower “log pass-through.”
  - Explained by same pass-through in levels across all retailers.

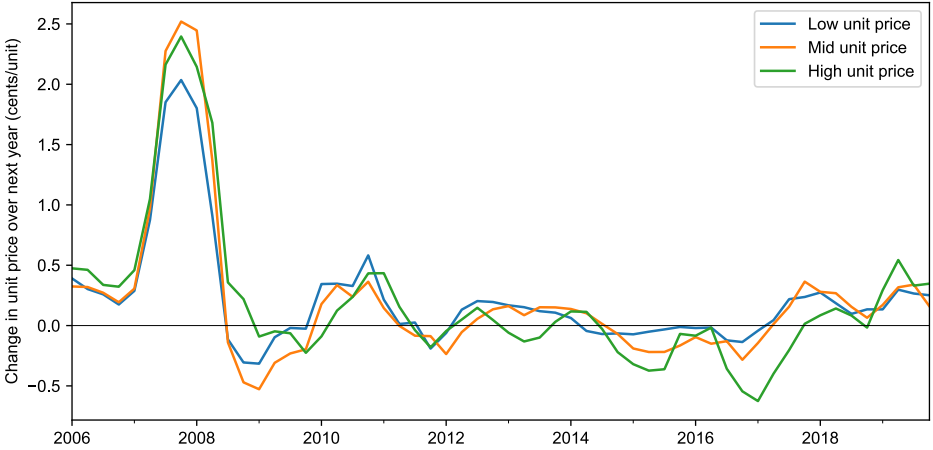
High-price products within category lower “log pass-through”

Figure: Inflation of rice products in NielsenIQ Retail Scanner data, split by unit price tercile.



Differences in pass-through disappear in absolute terms (levels)

Figure: Unit price change for rice products in NielsenIQ Retail Scanner data, split by unit price tercile.



All categories: High-price items have systematically lower "log pass-through"

$$\Delta \log p_{it} = \alpha_i + \beta_1 \Delta \log c_t + \sum_{g=2}^{3} \beta_g \left(1\{G(i,t) = g\} \times \Delta \log c_t\right) + \varepsilon_{it}.$$

Panel A: In percentages

	Retail price inflation		
	Rice	Flour	Coffee
Commodity Inflation × Mid Unit Price	−0.075**	−0.007	−0.064**
	(0.014)	(0.009)	(0.015)
Commodity Inflation × High Unit Price	−0.150**	−0.045**	−0.091**
	(0.022)	(0.009)	(0.017)
UPC FEs	Yes	Yes	Yes
N (thousands)	399.4	101.4	1570.0
R^2	0.15	0.05	0.14

- Higher unit price = larger gap btwn price and commodity cost (markup and/or non-commodity inputs).

All categories: No systematic differences in pass-through in levels

$$\Delta p_{it} = \alpha_i + \beta_1 \Delta c_t + \sum_{g=2}^{3} \beta_g \left(1\{G(i,t) = g\} \times \Delta c_t\right) + \varepsilon_{it}.$$

Panel B: In levels

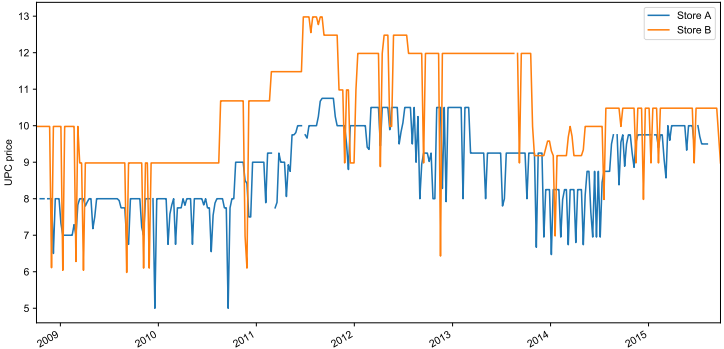
	Δ Retail price		
	Rice	Flour	Coffee
Δ Commodity Price × Mid Unit Price	0.059	0.027	−0.069
	(0.052)	(0.040)	(0.046)
Δ Commodity Price × High Unit Price	0.042	−0.067	−0.099*
	(0.100)	(0.044)	(0.058)
UPC FEs	Yes	Yes	Yes
N (thousands)	399.4	101.4	1570.0
R^2	0.07	0.05	0.14

- No systematic difference in pass-through in levels across unit price groups.

Exploiting variation in prices across retailers

- Consider two retailers selling the same UPC.
- If price differences partly due to variation in markups, then when the UPC cost rises, the high-markup retailer should increase price more in levels.

Figure: Prices of identical coffee UPC in two stores in same 3-digit ZIP code in Philadelphia, PA.



Exploiting variation in markups across retailers: Predictions

$$\Delta p_{ikt} = \beta^{\text{level}} \left(\Delta \bar{p}_{kt} \times \text{RelativePrice}_{ikt}\right) + \delta \text{RelativePrice}_{ikt} + \alpha_{kt} + \varepsilon_{ikt}, \tag{1}$$

$$\Delta \log p_{ikt} = \beta^{\log} \left(\Delta \log \bar{p}_{kt} \times \text{RelativePrice}_{ikt}\right) + \tilde{\delta} \text{RelativePrice}_{ikt} + \tilde{\alpha}_{kt} + \varepsilon_{ikt}, \tag{2}$$

- Δp_{ikt} is change in price of UPC k at retailer i from t to $t+4$.
- $\Delta \bar{p}_{kt}$ is average change in UPC k's price across all retailers.
- $\text{RelativePrice}_{ikt} = \log(p_{ikt}/\bar{p}_{kt})$ is retailer's price relative to others.

	Fixed percentage markup $p_i = \mu_i c$	Fixed additive markup $p_i = c + m_i$
Price change relative to average:		
Levels ($dp_i/d\bar{p}$)	$\approx 1 + \log(p_i/\bar{p})$	1
Logs ($d\log p_i/d\log\bar{p}$)	1	$\approx 1 - \log(p_i/\bar{p})$
Predicted interaction coefficient:		
Levels specification (1), β^{level}	1	0
Logs specification (2), $\beta^{\log}$	0	-1

Exploiting variation in markups across retailers

	Δ UPC Price (Δp_{ikt})			
	Rice	Flour	Coffee	All
	(1)	(2)	(3)	(4)
$\Delta \bar{p}_{kt} \times \text{RelativePrice}_{ikt}$	−0.022	0.058	0.192	−0.001
	(0.141)	(0.209)	(0.170)	(0.839)
UPC-Quarter FEs	Yes	Yes	Yes	Yes
N (millions)	0.399	0.101	1.570	100.4
R^2	0.46	0.48	0.51	0.59

- Result: $\beta^{\text{level}} \approx 0 \Rightarrow$ retailers change UPC price by same amount in *levels*.

Exploiting variation in markups across retailers

	Δ UPC Price (Δp_{ikt})				Δ Log UPC Price ($\Delta \log p_{ikt}$)			
	Rice	Flour	Coffee	All	Rice	Flour	Coffee	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \bar{p}_{kt} \times$ RelativePrice$_{ikt}$	−0.022	0.058	0.192	−0.001				
	(0.141)	(0.209)	(0.170)	(0.839)				
$\Delta \log \bar{p}_{kt} \times$ RelativePrice$_{ikt}$					−1.004**	−1.014**	−1.261**	−1.043**
					(0.125)	(0.184)	(0.084)	(0.073)
UPC-Quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N (millions)	0.399	0.101	1.570	100.4	0.399	0.101	1.570	100.4
R^2	0.46	0.48	0.51	0.59	0.62	0.61	0.58	0.61

- Result: $\beta^{\text{level}} \approx 0 \Rightarrow$ retailers change UPC price by same amount in *levels*.
- Makes "log pass-through" fall with markup. ($\beta^{\log} \approx -1$.)
- Holds for broader sample of nearly 1M UPCs across 235 retail chains!

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Explaining Pass-Through in Levels

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Manufacturing Industries

- NBER-CES Manufacturing Data: 459 four-digit SIC industries from 1958–2018.
  - Sales (shipments) and input costs (Census Annual Survey of Manufactures).
  - Price indices for output, materials, energy, labor (PPI data).

- Challenge: Price indices, not price levels! But with constant-returns, Leontief prod fn,

$$\rho^{\text{level}} \equiv \frac{\Delta p}{\Delta c} \qquad \Rightarrow \qquad \frac{\Delta p}{p} = \rho^{\text{level}} \left(\frac{cy}{py} \right) \frac{\Delta c}{c}.$$

  So we can estimate ρ^{level} using

$$\Delta \log p_{it} = \rho^{\text{level}} \left(\Delta \log c_{it} \times (\text{InputCosts}/\text{Sales})_{it-1} \right) + \varepsilon_{it}.$$

- Percentage markups imply $\rho^{\text{level}} = \mu > 1$, while additive markups imply $\rho^{\text{level}} \approx 1$.

Manufacturing Industries: Pass-Through

	Δ *Log Output Price*$_t$					
Inputs:	Materials		+ Energy		+ Production Labor	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log Input Price$_t$	0.690**	0.079	0.704**	0.005	0.796**	0.052
	(0.072)	(0.132)	(0.073)	(0.134)	(0.083)	(0.232)
$(\text{InputCost}/\text{Sales})_{t-1}$		0.004		0.008		0.023**
		(0.011)		(0.011)		(0.011)
Δ Log Input Price$_t \times (\text{InputCost}/\text{Sales})_{t-1}$		0.947**		1.041**		0.984**
		(0.203)		(0.201)		(0.286)
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	27 381	27 381	27 381	27 381	27 381	27 381
R^2	0.40	0.42	0.40	0.42	0.41	0.42

- Estimate $\rho^{\text{level}} \approx 1$ (complete pass-through in levels). Variation in InputCosts/Sales leads to incomplete log pass-through.
- Similar results with commodity price IV for input costs, horizons $1, \ldots, 5$ years.

Taking Stock of the Empirical Evidence

- Statistic: Long-run pass-through in levels of common cost shock ≈ 1.
- Retail gasoline. [in paper]
  - Complete pass-through in levels, explains variation in “log pass-through” across stations.
- Food products.
  - Complete pass-through in levels for nearly all products.
  - Explains variation in “log pass-through” across products within category.
  - Explains variation in “log pass-through” across retailers.
- Manufacturing industries.
  - Complete pass-through in levels across 400 industries spanning U.S. manufacturing.
- Previous literature. [in paper]
  - Survey of papers that estimate pass-through in levels of aggregate cost shocks.
  - Weight of evidence finds complete pass-through in levels of cost shocks, excise taxes.

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Explaining Pass-Through in Levels

Implications

Perfect Competition?

- One candidate explanation for pass-through in levels is perfect competition.
  - Gross markup $\mu = 1$, so complete pass-through in both levels and logs.
- However, perfect competition at odds with other features of the data.
  - Substantial price dispersion for identical products.
  - Sluggish price adjustment.
  - Finite firm-level demand elasticities.
  - Firms' prices elevated over available measures of cost.
- Price *dynamics* resemble perfect competition, but price *levels* don't.

Environment

- Set of goods partitioned into $J \geq 1$ *inside* goods and $K \geq 0$ *outside* goods.
- Prices for inside goods $\boldsymbol{p} = (p_1, \ldots, p_J)$. Outside goods vector $\boldsymbol{p}_0$.
- $D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ gives quantities demanded as function of prices, income Y.
- Each inside good produced with constant marginal cost c_j.

Environment

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- Prices for inside goods $\boldsymbol{p} = (p_1, \ldots, p_J)$. Outside goods vector $\boldsymbol{p}_0$.
- $D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ gives quantities demanded as function of prices, income Y.
- Each inside good produced with constant marginal cost c_j.
- **Assumption 1** (Nash-in-prices). For each $j \in \{1, \ldots, J\}$, price p_j set to maximize firm j's profits, taking all other prices and demand system D as given.
- **Assumption 2**. Given outside prices $\boldsymbol{p}_0$, income Y, and marginal costs $\boldsymbol{c}$, (1) an equilibrium exists (2) in which $\partial \log D_j / \partial \log p_j \in (-\infty, 0)$ for all j.

Sufficient Conditions For Pass-Through

- $D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ is scale invariant in $\boldsymbol{p}$ if there exists functions $\varphi_1, \ldots, \varphi_J$ s.t. for any $\lambda > 0$,

$$D_j(\lambda \boldsymbol{p}, \boldsymbol{p}_0, Y) = \lambda^{\varphi_j(\boldsymbol{p}, \boldsymbol{p}_0, Y, \lambda)} D_j(\boldsymbol{p}, \boldsymbol{p}_0, Y), \qquad \forall j.$$

  and $\partial \varphi_j / \partial p_j = 0$ for all j.

- $D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ is shift invariant in $\boldsymbol{p}$ if there exists functions $\psi_1, \ldots, \psi_J$ s.t. for any λ,

$$D_j(\boldsymbol{p} + \lambda, \boldsymbol{p}_0, Y) = (1 + \lambda \psi_j(\boldsymbol{p}, \boldsymbol{p}_0, Y, \lambda)) D_j(\boldsymbol{p}, \boldsymbol{p}_0, Y), \qquad \forall j.$$

  and $\partial \psi_j / \partial p_j = 0$ for all j.

Sufficient Conditions For Pass-Through

Proposition (Complete log pass-through)

Consider a shock that increases marginal costs by $d \log c_j = d \log c$ *for all* $j \in \{1, ..., J\}$*, holding fixed* $\boldsymbol{p}_0$ *and* Y*. If demand is scale invariant in* $\boldsymbol{p}$*, then* $d \log p_j = d \log c$ *for all* j*.*

Proposition (Complete pass-through in levels)

Consider a shock that increases marginal costs by $dc_j = dc$ *for all* $j \in \{1, ..., J\}$*, holding fixed* $\boldsymbol{p}_0$ *and* Y*. If demand is shift invariant in* $\boldsymbol{p}$*, then* $dp_j = dc$ *for all* j*.*

Sufficient Conditions For Pass-Through

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Proposition (Disjoint properties)

*If $D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ is scale invariant in $\boldsymbol{p}$, it is **not** shift invariant in $\boldsymbol{p}$, and vice versa.*

Scale Invariant Demand: Examples

Example (Homothetic preferences)

Suppose a representative consumer solves

$$\max_{\boldsymbol{q}} U(q_1, \ldots, q_J) \qquad \text{s.t.} \qquad \boldsymbol{p}'\boldsymbol{q} = Y.$$

If U is homothetic, then $\boldsymbol{q} = D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$, where D is scale invariant in $\boldsymbol{p}$, with $\boldsymbol{p}_0 = \emptyset$ and $\varphi_j = -1$ for all j.

- E.g., CES, Kimball, HSA demand.
- Incomplete pass-through of firm-specific shocks, but complete log pass-through of common shocks.

Scale Invariant Demand: Examples

Example (Nested homothetic preferences)

Suppose a continuum of industries $n \in [0,1]$, each with J firms. Rep. consumer maximizes

$$U = \left(\int_0^1 Q_n^{\frac{\sigma-1}{\sigma}} dn \right)^{\frac{\sigma}{\sigma-1}} \qquad \text{s.t.} \qquad \int_0^1 \sum_j p_{nj} q_{nj} dn = Y.$$

where $Q_n = Q_n(q_{n1}, \ldots, q_{nJ})$ is a homothetic aggregate over firms in industry n.

For each industry n, $\boldsymbol{q_n} = D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$, where D is scale invariant in $\boldsymbol{p} = (p_{n1}, \ldots, p_{nJ})$, with $\varphi_j = -\sigma$ for all j.

- E.g., nested CES (Atkeson and Burstein 2008), nested Kimball (Amiti et al. 2019).
- Complete log pass-through of industry-wide shocks.

Shift Invariant Demand: Examples

- If these demand systems are not consistent with pass-through in levels, which are?
- Answer: Quite a few shift invariant demand systems!

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Example (Log-linear demand)

Suppose the demand for good j follows

$$D_j(p_j, \boldsymbol{p}_0, Y) = \exp\left(a_j(\boldsymbol{p}_0, Y) - b_j(\boldsymbol{p}_0, Y)p_j\right).$$

Then D is shift invariant in p_j with $\psi_j = \left(\exp(-b_j\lambda) - 1\right)/\lambda$.

- Log-linear demand curves lead to pass-through in levels of firm-specific shocks.

  See e.g., Bulow and Pfleiderer (1983), Weyl and Fabinger (2013), Mrázová and Neary (2017).

Shift Invariant Demand: Examples

Example (Discrete choice with quasilinear preferences)

Suppose unit mass of consumers $i \in [0,1]$, each with income Y. Chooses one firm $j \in \{1, ..., J\}$

$$U_i = \max_j \{u_{ij}\} \qquad \text{s.t.} \qquad \begin{cases} u_{ij} = \delta_{ij} + q_{i0}, \\ p_j + p_0 q_{i0} = Y. \end{cases}$$

Demand system $D(\boldsymbol{p}, p_0, Y)$ from aggregating over all i is shift invariant in $\boldsymbol{p}$ with $\psi_j = 0$.

- E.g., logit, nested logit, mixed logit (Nevo 2001), multinomial probit.
- "Address models" like competition on a line (Hotelling 1929), unit circle (Salop 1979).
  - Market power derived from transport or search costs.

Takeaway: Class of Alternative Demand Systems Consistent with Evidence

- Large class of demand systems (but not homothetic demand) can match pass-through.
- **Neutrality:**
  - Any Marshallian demand system obeys $D(\lambda \boldsymbol{p}, \lambda \boldsymbol{p}_0, \lambda Y) = D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$.
  - Can further impose $D(\lambda \boldsymbol{p}, \lambda \boldsymbol{p}_0, Y) = D(\boldsymbol{p}, \boldsymbol{p}_0, Y)$ (not inconsistent with shift invariance).
- **Demand vs. conduct? Direct evidence on quantities:**
  - Homothetic demand predicts reallocation away from products with larger $\Delta \log p$.
  - Test using quantity shares of low vs. high-priced rice / flour / coffee.
  - No quantity response or if anything trading down.

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Simple Example

Empirical Evidence
- Food products
- Manufacturing industries

Explaining Pass-Through in Levels

Implications

Macro Implications

1. Asymmetry, size-dependence, and heterogeneity in “log pass-through.”
2. Mechanical effects of cost shocks on relative price and markup dispersion.
3. In industry model: Dynamics of industry gross margins, profits, and entry.
4. In multisector model: Differential pass-through of materials vs. labor costs. (Okun 1981)
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Asymmetry, size-dependence, and heterogeneity

- Suppose shift-invariant demand. Let $\mu_j = p_j / c_j$ be j's initial markup.
- In response to common cost shock dc, log pass-through $\rho_j^{\log}$ is

$$\rho_j^{\log} = \frac{\Delta \log p_j}{\Delta \log c_j} \approx \frac{1}{\mu_j} \left(1 + \frac{1}{2} \frac{\mu_j - 1}{\mu_j} \Delta \log c_j \right).$$

  - **Asymmetry**: Positive shocks appear to have higher pass-through: $\rho^{\log}(x) > \rho^{\log}(-x)$.
    E.g., Peltzman (2000), Bunn et al. (2024).
  - **Size-dependence**: Higher pass-through of large shocks: $\rho^{\log}(x_H) > \rho^{\log}(x_L)$ if $x_H > x_L$.
    Cavallo et al. (2024), Gagliardone et al. (2025).
  - **Heterogeneity**: If markups increase with firm size / quality, $\rho^{\log}$ falls with both.
    Size: Amiti et al. (2019), Gupta (2020). Quality: Chen and Juvenal (2016), Auer et al. (2018).

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  - Evidence from manufacturing: Cost increase leads to decline in log TFPR, LP dispersion.
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  - May help explain increased TFPR dispersion during recessions (Kehrig 2011).
- Δ price/markup dispersion typically interpreted as sign of Δ allocative efficiency.
  - Homothetic demand $\Rightarrow$ fall in markup of high-markup firms reallocates demand to them.
  - Data: cost shocks lower relative markup of expensive goods, but no reallocations to them!
  - Allocative efficiency interpretations hinge critically on homotheticity.

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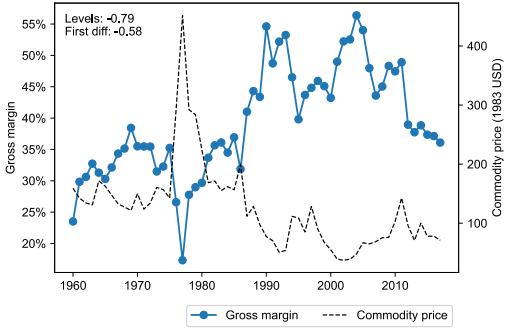
Industry model: Response of margins and entry to input cost changes

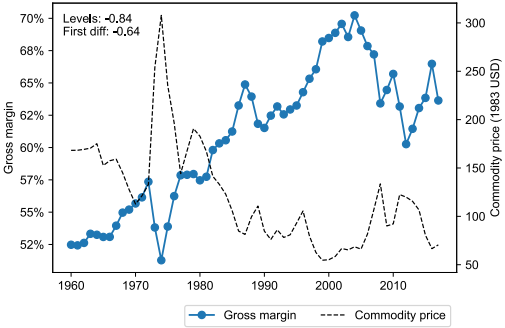
- Embed demand systems in standard monopolistic competition setup.
  - Mass N of symmetric firms, marginal cost c. Fixed entry cost f_e. Overhead cost f_o.
  - Aggregate industry demand inelastic, $Q = p^{-\theta}$, with $\theta < 1$.
  - Gross and operating margins = variable / operating profits as % of sales.
  - ζ is elasticity of entry to economic profits ($\zeta = 0$: fixed mass, $\zeta \to \infty$: free entry).

Response to cost increase, $dc > 0$			Gross margins dm^{gross}	Oper. margins dm^{op}	Mass of firms $d \log N$
Scale invariant	$\zeta = 0$	(Fixed mass)	0	> 0	0
	$\zeta \in (0, \infty)$		0	> 0	> 0
	$\zeta \to \infty$	(Free entry)	0	0	> 0
Shift invariant	$\zeta = 0$	(Fixed mass)	< 0	≤ 0	0
	$\zeta \in (0, \infty)$		< 0	≤ 0	≤ 0
	$\zeta \to \infty$	(Free entry)	< 0	0	≤ 0

Gross margins exhibit strong negative correlation with input costs



(a) Roasted coffee mfg vs. coffee.



(b) Bread, cake & related products mfg vs. wheat.

- In contrast with scale invariant demand, gross margins are not constant.
- Exhibit strong negative correlation with commodity input prices.

Shift invariant demand explains dynamics of gross, oper. margins, entry

Panel A: Retail Gasoline	Δ *Log Gross Margin*		Δ *Log Oper. Margin*		Δ *Log Num. Estabs*	
Source:	ARTS	IRS	ARTS	IRS	BDS	SUSB
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log Wholesale Price$_t$	−0.263**	−0.291**	0.490	0.125	−0.002	0.001
	(0.045)	(0.061)	(0.328)	(0.284)	(0.006)	(0.007)
N	39	26	15	26	39	24
R^2	0.54	0.49	0.20	0.01	0.00	0.00
Panel B: Manufacturing Industries	Δ *Log Gross Margin*		Δ *Log Oper. Margin*		Δ *Log Num. Estabs*	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Log Input Price$_t$	−0.188**	0.154	0.071	−0.086	0.007	−0.028
	(0.039)	(0.103)	(0.048)	(0.130)	(0.013)	(0.044)
Δ Log Input Price$_t$ × Inputs/Sales$_{t-1}$		−0.504**		0.270		0.049
		(0.188)		(0.221)		(0.063)
Industry FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	27 381	27 381	27 305	27 305	18 201	18 201
R^2	0.05	0.11	0.02	0.04	0.22	0.23

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Okun (1981) on materials costs

- Okun (1981) speculates a “special role for materials costs.”
  - “Some views of marking up direct costs distinguish increases in the costs of purchased materials from increases in standard unit labor costs, implying that the former are likely to be passed through to customers essentially on a dollars-and-cents basis, while the latter are passed through with a percentage markup.”

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- Continuum of sectors ω with identical firms, Leontief prod: $mc(\omega) = c + \beta(\omega)w$.
  - Consumers buy one unit from each sector $[0, \Omega]$.
  - Marginal utility of income given by prices in marginal sector, $p(\Omega)$.
- If demand is shift-invariant and marginal goods are labor-intensive $\beta(\Omega) \to 1$, then:
  - For material costs, $\rho_c = 1$, while for labor costs, $\rho_w \geq \mu(\omega)$.
  - The "adjusted" pass-through in levels of labor costs $\rho_w^{\text{adj}} = 1$, where

$$\frac{d \log p(\omega)}{d \log w} = \rho_w^{\text{adj}} \frac{\text{LaborCosts}(\omega) + \text{VariableProfits}(\omega)}{\text{Sales}(\omega)}.$$

Revisiting Okun (1981): Empirical evidence

	Δ Log Output Price$_t$		
	(1)	(2)	(3)
	Cost Shares	Sales Shares	Sales Shares
Δ Log Material Price$_t$ $\times$ Material Share$_{t-1}$	0.798**	1.046**	
	(0.085)	(0.108)	
Δ Log Energy Price$_t$ $\times$ Energy Share$_{t-1}$	0.715**	0.859**	
	(0.271)	(0.378)	
Δ Log Production Wage$_t$ $\times$ Labor Share$_{t-1}$	1.070**	2.095**	
	(0.217)	(0.281)	
Δ Log Production Wage$_t$ $\times$ (Labor + Variable Profits) Share$_{t-1}$			
Industry FEs	Yes	Yes	
Year FEs	Yes	Yes	
N	27 374	27 374	
R^2	0.49	0.50	

- Labor costs appear to have complete log pass-through, $>$1:1 pass-through in levels.

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	(0.271)	(0.378)	(0.362)
Δ Log Production Wage$_t$ $\times$ Labor Share$_{t-1}$	1.070**	2.095**	
	(0.217)	(0.281)	
Δ Log Production Wage$_t$ $\times$ (Labor + Variable Profits) Share$_{t-1}$			0.951**
			(0.209)
Industry FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
N	27374	27374	27374
R^2	0.49	0.50	0.49

- Labor costs appear to have complete log pass-through, $>$1:1 pass-through in levels.
- Corrected when we account for the fact that additive markup priced relative to wage.

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Low volatility of consumer inflation

- Input-output model with 405 BEA industries, Calvo frictions á la Rubbo (2023).
- Take prices of upstream ("Stage 1") industries from data.
- With fixed percentage markups, volatility of consumer price inflation is too high.

	Std. deviation of annual inflation, 1982–2018	Cost-weighted average markup	
		Mfg.	All
Personal Consumption Expenditures (PCE) Price Index	1.1%		
Consumer Price Index for All Urban Consumers	1.3%		
Scale invariant, Variable costs (VC) = Materials	2.5%	1.48	2.19

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Scale invariant, VC = Materials + wages	1.7%	1.19	1.31
Scale invariant, VC = Materials + wages + cons. of fixed capital	1.6%	1.11	1.17

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- Additive markups reconciles low volatility of inflation with large markups.

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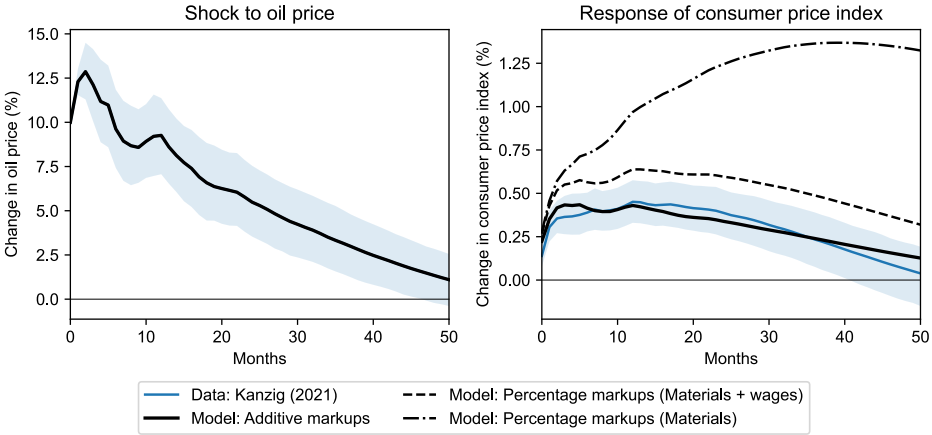
Shift invariant, Additive markups	1.3%	1.0–1.5	1.0–2.2

Response of consumer inflation to oil supply shock



- *Note:* Impulse responses to an oil supply news shock from Känzig (2021). The solid line is the point estimate and the shaded areas are 68 percent confidence bands. The changes in the oil price in the left panel are treated as exogenous changes in the oil price in the model. The model lines in the right panel plot the response of the personal consumption expenditures (PCE) price index in the calibrated models given the changes in oil prices.

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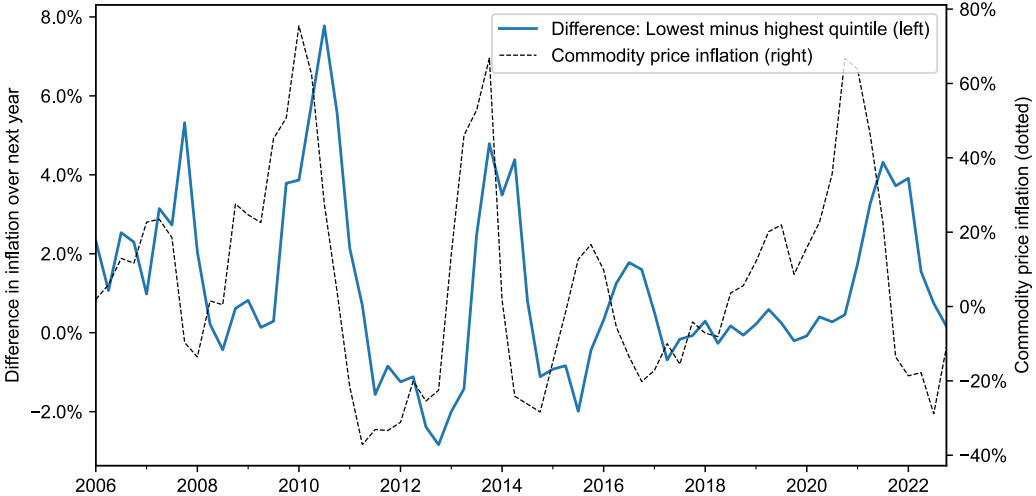
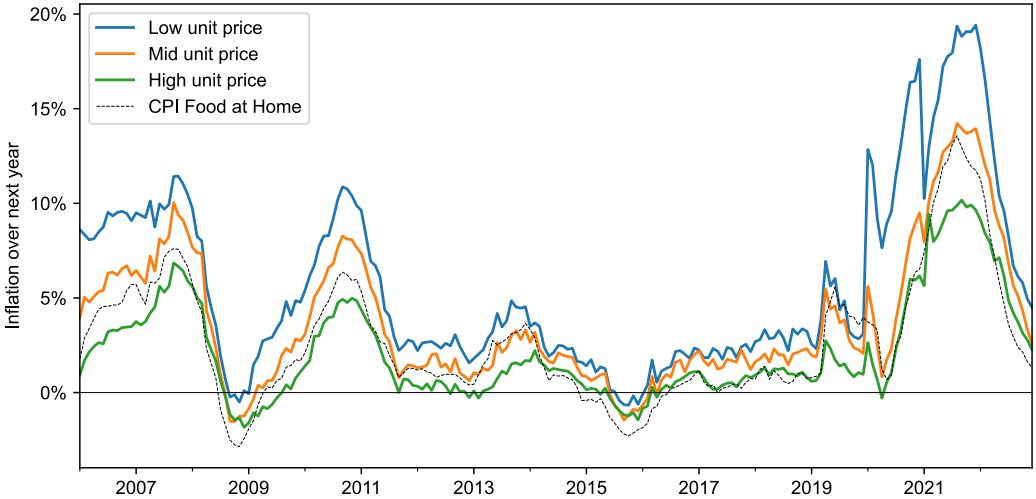
Inflation inequality moves with commodity costs within category (e.g. coffee)

- **Coffee**: Same $\Delta p \rightarrow$ higher inflation for low-price varieties (bought by low-inc hh's).



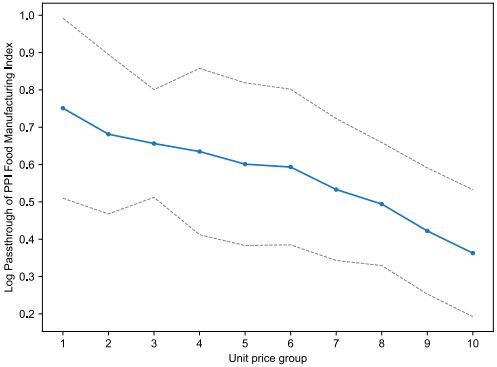
“Cheapflation cycles” across entire food-at-home bundle



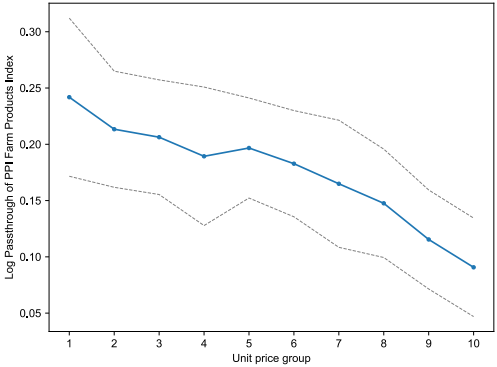
Can pass-through in levels explain recent cheapflation / inflation inequality?

- Approach: split each product category into 10 groups by relative price.
- Fit pass-through of upstream costs to prices over 2006–2019, then extrapolate using 2020–2023 increase in costs.

(a) Log pass-through of Food Manufacturing PPI.

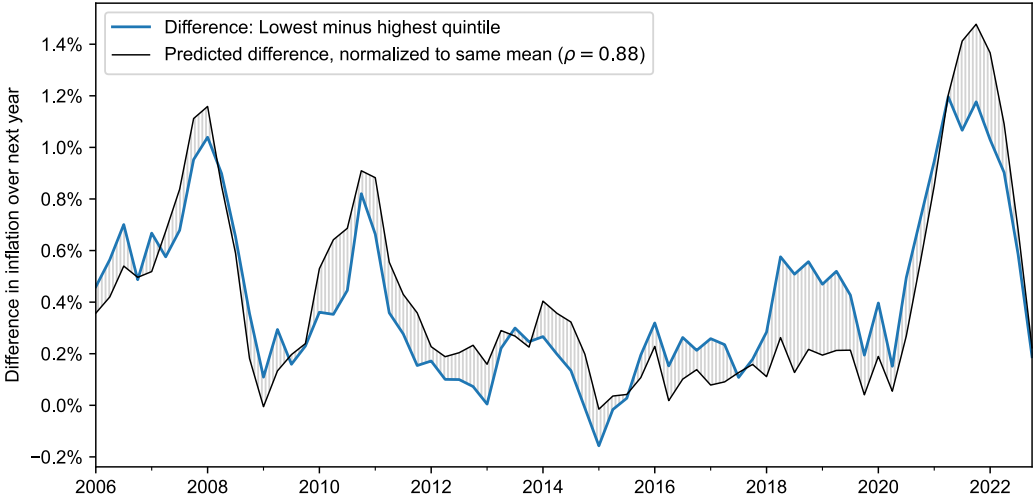


(b) Log pass-through of Farm Products PPI.



Can pass-through in levels explain recent cheapflation / inflation inequality?

- Fully accounts for 2021–2023 surge!



Conclusion

- Empirical evidence: Pass-through of common cost shocks is complete in *levels*.
- Not consistent with homothetic demand in workhorse models.
- Shift-invariant demand systems can explain pass-through behavior.
- Pass-through in levels rationalizes several other facts:
  - Incomplete log pass-through + asymmetry, size-dependence, heterogeneity.
  - Fluctuations in relative price / markup dispersion, without changes in efficiency.
  - Dynamics of industry profits, margins, and entry.
  - Differential pass-through of labor vs. materials costs.
  - Low volatility of consumer price inflation.
  - Cycles in cheapflation and inflation inequality.

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